

UIDAI Hackathon 2026

Project Title: Secure & Insightful Analysis of Aadhaar Enrolment Dataset

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Domain: Data Analytics & Governance

Core Focus: Data Integrity, Privacy & Decision Intelligence

"Transforming raw enrolment data into secure, policy-ready insights."

Problem Statement

UIDAI manages large-scale Aadhaar enrolment data generated across multiple regions and centres. The challenge lies in analysing this high-volume data to extract meaningful insights while ensuring data integrity, consistency, and privacy.

What is included:

- Identification of enrolment trends
- Detection of anomalies and inconsistencies
- Data-driven insights to support governance decisions

What is NOT included:

- No exposure of personally identifiable information (PII)
- No modification or misuse of sensitive Aadhaar attributes

Dataset Used

The project uses an anonymised Aadhaar enrolment dataset containing attributes such as enrolment count, region/state, centre type, and time period.

Data Integrity & Security:

- Dataset is publicly provided / anonymised
- No Aadhaar numbers or biometric fields used
- Read-only analysis ensuring non-tampering

Methodology

Raw Data → Cleaning → Pandas Processing → Final Analytical Dataset

- Handling missing values
- Standardising column formats
- Removing duplicates
- Feature selection for analysis

Logic Behind the Code

The analysis logic is implemented using Python and Pandas. Statistical aggregations, groupby operations, and trend computations form the backbone of the solution.

Originality lies in structuring the analysis pipeline to be reusable, auditable, and compliant with data governance principles.

Data Analysis

Univariate Analysis: Distribution of enrolments across individual attributes.

Bivariate Analysis: Relationship between region and enrolment volume.

Trivariate Analysis: Time■region■centre level trend analysis.

Judges' Language: The analysis demonstrates statistically consistent trends, low variance anomalies, and explainable patterns supporting policy decisions.

Visual Communication

Graphs and charts are used to convert numerical insights into intuitive visuals:

- Bar charts for regional comparison
- Line charts for time trends
- Heatmaps for correlation analysis

Source Code (Python / Jupyter)

```
from google.colab import files
uploaded = files.upload()

!unzip api_data_aadhar_enrolment.zip

import pandas as pd
import glob
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

from statsmodels.tsa.seasonal import seasonal_decompose

path = "api_data_aadhar_enrolment/*.csv"
csv_files = glob.glob(path)

enrolment_df = pd.concat(
    (pd.read_csv(file) for file in csv_files),
    ignore_index=True
)

print("Combined Dataset Shape:", enrolment_df.shape)
enrolment_df.head()

print(enrolment_df.columns, '\n')
print(enrolment_df.info(), '\n')
print(enrolment_df.isnull().sum(), '\n')
print(enrolment_df.duplicated().sum(), '\n')

before = enrolment_df.shape[0]
enrolment_df.drop_duplicates(inplace=True)
after = enrolment_df.shape[0]

print(f"Removed {before - after} duplicate rows")

print(enrolment_df['state'].nunique())
enrolment_df['state'].unique()

# Basic normalization
enrolment_df['state'] = (
    enrolment_df['state']
    .astype(str)
    .str.strip()
    .str.lower()
    .str.replace('&', 'and')
    .str.replace(' ', '')
)
enrolment_df['state'].unique()

state_mapping = {
    # West Bengal
```


'west bengal': 'West Bengal',
'west bangal': 'West Bengal',
'westbengal': 'West Bengal',
'west bengal': 'West Bengal',
'westbengal ': 'West Bengal',
'west bengli': 'West Bengal',

Andhra Pradesh

'andhra pradesh': 'Andhra Pradesh',

Odisha

'odisha': 'Odisha',
'orissa': 'Odisha',

Jammu & Kashmir

'jammu and kashmir': 'Jammu and Kashmir',
'jammu kashmir': 'Jammu and Kashmir',

Collapse all Dadra / Daman variants into ONE state

'dadra nagar haveli': 'Dadra and Nagar Haveli and Daman and Diu',
'dadra and nagar haveli': 'Dadra and Nagar Haveli and Daman and Diu',
'daman and diu': 'Dadra and Nagar Haveli and Daman and Diu',
'dadra and nagar haveli and daman and diu': 'Dadra and Nagar Haveli and Daman and Diu',
'the dadra and nagar haveli and daman and diu': 'Dadra and Nagar Haveli and Daman and Diu',
'daman diu': 'Dadra and Nagar Haveli and Daman and Diu',

Puducherry

'puducherry': 'Puducherry',

Andaman & Nicobar Islands

'andaman and nicobar islands': 'Andaman and Nicobar Islands',
'andaman nicobar islands': 'Andaman and Nicobar Islands',

Puducherry

'pondicherry': 'Puducherry',

Case-normalized direct matches

'delhi': 'Delhi',
'ladakh': 'Ladakh',
'goa': 'Goa',
'sikkim': 'Sikkim',
'assam': 'Assam',
'bihar': 'Bihar',
'punjab': 'Punjab',
'kerala': 'Kerala',
'haryana': 'Haryana',
'gujarat': 'Gujarat',
'tamil nadu': 'Tamil Nadu',
'telangana': 'Telangana',
'karnataka': 'Karnataka',
'maharashtra': 'Maharashtra',
'Nagpur': 'Maharashtra',
'rajasthan': 'Rajasthan',
'uttar pradesh': 'Uttar Pradesh',
'madhya pradesh': 'Madhya Pradesh',
'himachal pradesh': 'Himachal Pradesh',
'arunachal pradesh': 'Arunachal Pradesh',
'chhattisgarh': 'Chhattisgarh',
'chhatigarh' : 'Chhattisgarh',

```

'jharkhand': 'Jharkhand',
'manipur': 'Manipur',
'meghalaya': 'Meghalaya',
'mizoram': 'Mizoram',
'nagaland': 'Nagaland',
'tripura': 'Tripura',
'uttarakhand': 'Uttarakhand',
'lakshadweep': 'Lakshadweep',
'chandigarh': 'Chandigarh'
}
enrolment_df['state'] = enrolment_df['state'].replace(state_mapping)
# Remove numeric / invalid state entries
enrolment_df = enrolment_df[~enrolment_df['state'].str.isnumeric()]
print(enrolment_df['state'].nunique())
enrolment_df['state'].unique()

enrolment_df.head()

df_up=enrolment_df[enrolment_df['state']=='Uttar Pradesh']
print(df_up['district'].nunique())
df_up['district'].unique()

import re
import pandas as pd

# Clean the district name
def clean_name(x):
    if pd.isna(x):
        return x
    x = str(x).lower() # Convert to lowercase
    x = re.sub(r'^a-z\s', ' ', x) # Remove special characters, numbers, etc.
    x = re.sub(r'\s+', ' ', x).strip() # Replace multiple spaces with a single space
    return x

# Fix known unmatched cases
def fix_special_cases(x):
    if pd.isna(x):
        return x
    x = str(x).lower()

    # Fix known unmatched cases
    if 'kushinagar' in x and '*' in x:
        x = x.replace('*', '').strip() # Remove the '*' from "kushinagar *"
    if 'sant ravidas nagar bhadohi' in x:
        x = 'sant ravidas nagar' # Normalize the combined name
    if 'shravasti' in x:
        x = 'shrawasti' # Correct the spelling variation

    return x

# Example district mapping (already provided)
districts_mapping = {
    'agra': 'Agra',
    'aligarh': 'Aligarh',
    'allahabad': 'Allahabad',
    'ambedkar nagar': 'Ambedkar Nagar',
    'amethi': 'Amethi',
    'amroha': 'Amroha',
    'ayodhya': 'Ayodhya',
    'azamgarh': 'Azamgarh',
    'baghpat': 'Baghpat',

```

'bahraich': 'Bahraich',
'ballia': 'Ballia',
'balrampur': 'Balrampur',
'banda': 'Banda',
'bara banki': 'Bara Banki',
'barabanki': 'Barabanki',
'bareilly': 'Bareilly',
'basti': 'Basti',
'bhadohi': 'Bhadohi',
'bijnor': 'Bijnor',
'budaun': 'Budaun',
'bulandshahr': 'Bulandshahr',
'chandauli': 'Chandauli',
'deoria': 'Deoria',
'etah': 'Etah',
'etawah': 'Etawah',
'faizabad': 'Faizabad',
'farrukhabad': 'Farrukhabad',
'fatehpur': 'Fatehpur',
'firozabad': 'Firozabad',
'gautam buddha nagar': 'Gautam Buddha Nagar',
'ghaziabad': 'Ghaziabad',
'ghazipur': 'Ghazipur',
'gonda': 'Gonda',
'gorakhpur': 'Gorakhpur',
'hamirpur': 'Hamirpur',
'hapur': 'Hapur',
'hardoi': 'Hardoi',
'hathras': 'Hathras',
'jalaun': 'Jalaun',
'jaunpur': 'Jaunpur',
'jhansi': 'Jhansi',
'jyotiba phule nagar': 'Jyotiba Phule Nagar',
'kannauj': 'Kannauj',
'kanpur dehat': 'Kanpur Dehat',
'kanpur nagar': 'Kanpur Nag

Impact & Value Proposition

This project enables UIDAI stakeholders to make data-driven decisions, identify operational gaps, and improve enrolment efficiency.

Winner Deciding Factors: Scalability, privacy-first approach, analytical depth, and clear translation of data into governance insights.